



CONTEXTWITHRAG

Project documentation - AI assistant for source code, requirements and project documents

Final year project

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Stack: FastAPI, MongoDB, ChromaDB, LangChain, Google Gemini, HTML/CSS/JavaScript

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1. Project overview

CONTEXTWITHRAG is a web application that lets a user upload project files (source code, PDF, Word and PowerPoint documents) and then ask questions about them in a chat. The answers are produced by Google Gemini, but only after the application has searched the user's own files for the most relevant passages. This technique is called Retrieval-Augmented Generation (RAG): the model answers from the retrieved context instead of guessing.

Each user has a private account. Uploaded files, chats and memory are stored per user and are never visible to another user.

What problem it solves

- Reading a large codebase or a long requirements document takes time. The assistant finds the relevant part and explains it.
- Answers show which files they were based on, so they can be checked.
- The assistant remembers earlier conversations, so follow-up questions work across chats.

2. Features

Feature	Description
Accounts	Register and login with username and password. Passwords are stored as bcrypt hashes. Password rules: 8+ characters, uppercase, lowercase, digit and symbol.
Secure sessions	On login the server creates a random token and stores only its SHA-256 hash in MongoDB. Logout deletes it immediately. Default lifetime is 720 minutes.
Knowledge base	Upload code files, PDF, DOCX and PPTX. Files are split into chunks and stored as vectors in ChromaDB. Maximum size is 5 MB per file.
Chat with your project	Ask a question; the 8 most relevant chunks are sent to Gemini with the question. The reply lists the source files used.
Short-term memory	The last 10 messages of the current chat are sent with each question. Chats expire from MongoDB after 24 hours.
Long-term memory	Every question/answer pair is stored in ChromaDB and recalled in later chats when it is related to the new question. The user can clear it.
Dashboard cards	The welcome screen shows files indexed, recent chats and remembered exchanges.
Search	The header search box filters the chat list and file list (Ctrl+K).
Responsive UI	Fixed header and sidebar on desktop, slide-out drawer on phones and tablets.

3. Technology stack

Layer	Technology	Used for
Backend	Python 3.12, FastAPI, Uvicorn	REST API and serving the web page
Document loading	LangChain (langchain-community, langchain-text-splitters), PyMuPDF, python-docx, python-pptx	Reading files page by page and splitting them into chunks
Vector database	ChromaDB (persistent, on disk)	Document chunks and long-term memory, searched by meaning
Embeddings	all-MiniLM-L6-v2 (ONNX, built into ChromaDB)	Turning text into vectors; runs locally, no API key needed
Database	MongoDB (pymongo)	Users, login sessions, file list, chats and messages
Language model	Google Gemini (google-genai), default model gemini-2.5-flash	Writing the answers
Security	bcrypt, HTTP Bearer tokens	Password hashing and authentication
Frontend	HTML, CSS and JavaScript in one file (index.html)	The whole user interface
Frontend libraries	marked, DOMPurify, Font Awesome 6.5, Inter font	Markdown rendering, HTML sanitising, icons, typography

4. How it works

4.1 Uploading a file

1. The browser checks the file size. A file larger than 5 MB is rejected with a pop-up message and is not sent.
2. The server checks the file type and size again.
3. The file is loaded with LangChain. A PDF is read page by page with PyMuPDF, a PowerPoint slide by slide, and a Word file including its tables.
4. The text is split into chunks of about 1200 characters with 200 characters of overlap. Text is cut at paragraph and sentence boundaries; source code is cut at function and class boundaries.
5. Each chunk is stored in the ChromaDB collection `documents` with the user id, file id, file name and page number.
6. The file name, size and chunk count are saved in the MongoDB collection `files`.

4.2 Asking a question

1. The question is sent to `POST /api/chat` with the login token.
2. Short-term memory: the last 10 messages of this chat are read from MongoDB.
3. Document search: ChromaDB returns the 8 chunks of this user's files that are closest in meaning to the question.
4. Long-term memory: ChromaDB returns up to 3 related question/answer pairs from the user's other chats (only if the cosine distance is 0.75 or less).
5. A prompt is built from four parts: long-term memory, document context (with file name and page), recent conversation, and the question.
6. Gemini generates the answer. A busy server (503) is retried up to 3 times; if the configured model is not available another Gemini text model is used.
7. The question and answer are saved in MongoDB and the pair is added to long-term memory.
8. The browser shows the answer in Markdown with the list of referenced files.

4.3 Two kinds of memory

	Short-term memory	Long-term memory
Stored in	MongoDB (sessions, messages)	ChromaDB (long_term_memory)
Contains	The messages of each chat	Question/answer pairs from all chats
Found by	Chat id, newest first	Similarity of meaning
Lifetime	24 hours (automatic TTL index)	Until the user clicks Clear

5. Project structure

File or folder	Purpose
index.html	The complete frontend: login flow, header, sidebar, chat, styles and JavaScript
backend/main.py	Creates the FastAPI app, connects routes, serves index.html and the /img folder
backend/config.py	Reads settings from the .env file
backend/database.py	MongoDB connection, collections and indexes
backend/security.py	Password hashing, login tokens, current-user check
backend/schemas.py	Request and response models with validation (for example the password rules)
backend/routes_auth.py	Register, login, logout and current user
backend/routes_files.py	Upload, list and delete files
backend/routes_chat.py	Chat, chat history and memory endpoints; builds the prompt
backend/processor.py	Loads files with LangChain and splits them into chunks
backend/vector_store.py	All ChromaDB operations: add, search, delete, remember, recall
backend/llm.py	Google Gemini client with retry and model fallback
img/	Header slideshow images (slide0.webp to slide3.webp) and the original PNG banners
chroma_data/	ChromaDB files on disk (created automatically, not committed to git)
view_chroma.py	Read-only tool to look inside ChromaDB from the terminal
requirements.txt	Python packages
.env / .env.example	Settings and secrets / template without secrets

6. Setup and run commands

All commands are run in a terminal (PowerShell) inside the project folder.

6.1 Requirements

- Python 3.12
- A MongoDB database (local MongoDB or MongoDB Atlas)
- A Google Gemini API key from <https://aistudio.google.com/apikey>
- Internet access (Gemini, and Atlas if used). The first upload also downloads the embedding model once.

6.2 First-time setup

Go to the project folder:

```
cd "C:/Users/hites/Downloads/PROJECT FINAL YEAR"
```

Create the virtual environment (only if the `.venv` folder does not exist):

```
python -m venv .venv
```

Install the packages:

```
.venv/Scripts/python.exe -m pip install -r requirements.txt
```

Create the settings file from the template, then open `.env` and fill in `GOOGLE_API_KEY` and `MONGO_URI`:

```
copy .env.example .env
```

6.3 Start the server

```
.venv/Scripts/python.exe -m uvicorn backend.main:app --port 8000
```

Open <http://localhost:8000> in the browser. Stop the server with Ctrl+C.

While developing, add `--reload` at the end of the same command so the server restarts when a Python file changes.

Changes to `index.html` do not need a restart; refresh the browser. Changes to Python files or to `.env` need a restart (or `--reload`).

6.4 Useful checks

Health check (also shows whether the Gemini key is set and the upload limit):

```
curl http://localhost:8000/api/health
```

Interactive API documentation generated by FastAPI is available at <http://localhost:8000/docs> while the server is running.

7. Configuration (.env)

Setting	Default	Meaning
GOOGLE_API_KEY	(empty)	Gemini API key. Chat returns an error until it is set.
GEMINI_MODEL	gemini-2.5-flash	Model used for answers
LOGIN_EXPIRE_MINUTES	720	How long a login stays valid
MONGO_URI	mongodb://localhost:27017	MongoDB connection string
MONGO_DB	nexus_ai	Database name (kept from the project's earlier name so existing data still works)
SHORT_TERM_TTL_HOURS	24	Chats are deleted after this many hours
SHORT_TERM_TURNS	10	Recent messages sent to the model
CHROMA_DIR	chroma_data	Folder for ChromaDB files
MAX_UPLOAD_MB	5	Maximum size of one uploaded file

The .env file contains secrets (API key, database password). It is listed in .gitignore and must not be shared or uploaded.

8. API reference

All endpoints except register, login and health need the header `Authorization: Bearer <token>`.

Method	Path	Description
POST	/api/auth/register	Create an account; returns a token and the user
POST	/api/auth/login	Login; returns a token and the user
POST	/api/auth/logout	Invalidate the current token
GET	/api/auth/me	The logged-in user
POST	/api/files	Upload one or more files (form field: uploads)
GET	/api/files	List the user's indexed files
DELETE	/api/files/{file_id}	Remove a file and its chunks
POST	/api/chat	Ask a question (message, optional session_id); returns answer and sources
GET	/api/sessions	List recent chats
GET	/api/sessions/{session_id}/messages	Messages of one chat
DELETE	/api/sessions/{session_id}	Delete a chat
GET	/api/memory	Number of remembered exchanges and the chat lifetime
DELETE	/api/memory	Clear the user's long-term memory
GET	/api/health	Server status, Gemini key present, upload limit
GET	/	The web application (index.html)
GET	/img/...	Header slideshow images

Supported file types

Documents: pdf, docx, pptx. Text and code: txt, md, rst, csv, json, xml, yaml, yml, toml, ini, cfg, env, log, sql, py, js, jsx, ts, tsx, java, kt, c, h, cpp, hpp, cs, go, rs, rb, php, swift, scala, dart, r, sh, bat, ps1, html, htm, css, scss, vue, svelte, ipynb.

9. Databases and how to view the data

9.1 MongoDB collections

Collection	Contents
users	username, email, password hash, active flag, creation date
login_sessions	token hash, user id, expiry time (removed automatically when expired)
files	file name, size, number of chunks, owner, upload date
sessions	chats: title, owner, last update (expire after 24 hours)
messages	chat messages with role, content and sources (expire after 24 hours)

View this data with MongoDB Compass or the MongoDB Atlas website, using the same connection string as in `.env`.

9.2 ChromaDB collections

Collection	Contents
documents	Chunks of uploaded files with user id, file id, file name and page number
long_term_memory	Question/answer pairs with user id, chat id and date

ChromaDB has no viewer of its own, so the project includes `view_chroma.py` (read-only).

Summary of both collections:

```
.venv/Scripts/python.exe view_chroma.py
```

List document chunks, or remembered exchanges:

```
.venv/Scripts/python.exe view_chroma.py docs
```

```
.venv/Scripts/python.exe view_chroma.py memory
```

Search the chunks by meaning (a smaller distance means a closer match):

```
.venv/Scripts/python.exe view_chroma.py search "invoice total"
```

Add a number at the end to change how many rows are shown, for example `docs 30`.

10. Using the application

10.1 Login flow

1. Welcome: drag the slider to the right (or press Enter on it).
2. Sign in: choose Login or Register, fill in the form and submit. The Register form shows live password rules.
3. Application: all three steps turn green, a welcome message with a progress bar is shown, and the app opens.

10.2 Main screen

Area	What it does
Header	Name, image slideshow, search box and the current chat chip
Sidebar - Main	New chat, Upload files
Sidebar - Knowledge base	Drop zone and the list of indexed files (remove with the x button)
Sidebar - Recent chats	Open or delete an earlier chat
Sidebar - Long-term memory	Number of remembered exchanges and the Clear button
Content area	Dashboard cards and welcome panel for a new chat; otherwise the conversation
Composer	Type a question and press Enter or Analyze (Shift+Enter for a new line)

10.3 Keyboard shortcuts

Shortcut	Action
Ctrl + K	Focus the search box
Ctrl + B	Collapse or expand the sidebar (desktop)
Enter	Send the question
Shift + Enter	New line in the question
Esc	Close a confirmation dialog

11. Work completed in this build

Area	What was done
User interface	Complete redesign as a light dashboard: design tokens, fixed gradient header, sidebar with icon-rail mode, scrolling content area, cards, tables, buttons, badges, inputs, confirmation dialog and toasts. Desktop is drawn at 80% scale (90% on very wide screens).
Admin removed	The user-management screen, the admin API routes, role checks and the role field were removed. Every account is a normal user.
Header slideshow	Four banner images added with crossfade, slow zoom, caption chips and progress dots. Light WebP copies (50-90 KB) are used; the server serves them from /img.
Login	White background, three-step flow (Welcome, Sign in, Application), swipe slider, input icons, success screen with progress bar and confetti.
Register form	Two-column layout that fits the screen without scrolling; the background no longer moves.
PDF analysis	Switched from plain pypdf extraction to LangChain with PyMuPDF. Page numbers are kept, text is split at natural boundaries, code is split by function and class, and 8 chunks (previously 4) are retrieved per question.
Upload limit	5 MB per file for every file type, checked in the browser (pop-up message) and on the server.
Name	The application was renamed from Nexus AI to CONTEXTWITHRAG.
Tools	view_chroma.py added to inspect ChromaDB data.

How it was verified

- Pages were opened in a browser at desktop, laptop and phone sizes; there were no console errors.
- PDF loading, chunking, page-aware search and deletion were tested with a generated three-page PDF against a temporary ChromaDB store.
- After a server restart the page, the images and the health endpoint returned 200 and the old admin route returned 404.
- Not verified by an automated test: a full login and file upload through the browser with a real account. This should be checked by hand once.

12. Known limits and future work

Limit	Detail	Possible improvement
Scanned PDFs	A PDF that is only an image has no text and is rejected.	Add OCR
Whole-document questions	Only the 8 closest chunks reach the model, so 'total of all rows' over a long file can be wrong.	Send the full text for small files on aggregate questions
Tables in PDFs	Tables are read as plain text; columns can be mixed up.	Extract tables with their rows and columns
Arithmetic	The model calculates by itself and can make mistakes with many or large numbers.	Enable Gemini code execution for exact results
Files indexed earlier	Files uploaded before the LangChain change keep the old chunks without page numbers.	Remove and upload them again
Chat lifetime	Chats disappear after 24 hours by design.	Change SHORT_TERM_TTL_HOURS
Text size	The compact design uses small text on desktop.	Raise --app-zoom in index.html
Open CORS	The API accepts requests from any origin, which is convenient for development.	Restrict origins before deploying publicly

13. Troubleshooting

Problem	Cause and fix
Server does not start: cannot connect to MongoDB	Check MONGO_URI in .env, the internet connection, and that your IP address is allowed in MongoDB Atlas (Network Access).
Port 8000 is already in use	Another copy of the server is running. Stop it with Ctrl+C in its terminal, or start on another port with --port 8001.
Chat says GOOGLE_API_KEY is not configured	Add the key to .env and restart the server.
Gemini API error 429 or 503	Rate limit or busy service. Wait a moment and ask again.
Header images are missing	The img folder must contain slide0.webp to slide3.webp, and the server must have been restarted after they were added.
File is skipped: larger than 5 MB	Reduce the file or raise MAX_UPLOAD_MB in .env and restart.
File is skipped: no readable text found	The PDF is scanned (image only). Use a PDF with selectable text.
Session expired message	The login is older than LOGIN_EXPIRE_MINUTES. Login again.
First upload is slow	ChromaDB downloads the embedding model once; later uploads are fast.
Page looks unchanged after an update	Refresh the browser with Ctrl+F5.